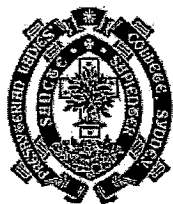


Student's name

Student's number

Teacher's name



PLC PRESBYTERIAN
LADIES' COLLEGE
SYDNEY
1888

2008
TRIAL
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total Marks – 84

- Attempt questions 1-7
- All questions are of equal value

1	2	3	4	5	6	7	Total	Total
							/84	%

Question 1 (12 marks) Start a new sheet of writing paper. Marks

- (a) Evaluate, $\lim_{x \rightarrow 0} \frac{3 \sin \frac{x}{2}}{x}$, showing all working. 2
- (b) Find the coordinates of the point, P , that divides the interval AB internally in the ratio of 4:5 if $A(-2,3)$ and $B(1,0)$. 2
- (c) Find k if $x^{2k+3} = e^{9 \ln x}$, where $x > 0$. 2
- (d) $\int \cos^2 4x \, dx$ 3
- (e) Use the substitution $u = 1 + x^5$ to evaluate $\int_{-1}^1 x^4 \sqrt{1+x^5} \, dx$ 3

End of Question 1

Question 2 (12 marks) Start a new sheet of writing paper. Marks

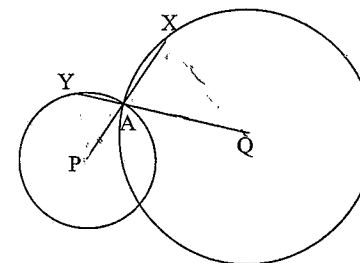
- (a) Solve $\frac{2}{x-1} \geq \frac{3}{x}$, $x \neq 0$, $x \neq 1$ 3
- (b) Given two roots of $2x^3 - kx + 8 = 0$ are equal, find k . 3
- (c) Find the constant term in $(x^3 - \frac{1}{x})^8$ 3
- (d) A particle is moving in a straight line with its acceleration as a function of x given by $\ddot{x} = -8x^3$. It is initially at the origin and is travelling with a velocity of 4m/s .
- i. Find the maximum speed of the particle. 1
- ii. Show that $\dot{x} = 2\sqrt{4-x^4}$ 2

End of Question 2

Question 3 (12 marks) Start a new sheet of writing paper. Marks

- (a) i. Show that $\cos 3x = 4\cos^3 x - 3\cos x$ 2
- ii. Hence, or otherwise, find $\int \cos x \sin^2 x \, dx$ 2
- (b) An oven has been heated to a constant temperature of 180°C . A cake mixture, with a temperature of 20°C is placed in the oven and after 15 minutes its temperature is measured at 100°C . The heating rate is proportional to the difference between the cake temperature and the oven temperature.
- i. Show that the equation for the cake temperature is given by $T = 180 - 160e^{-0.046t}$. 2
- ii. What will be the temperature of the cake after 30 minutes? 1
- iii. How long will it take for the cake to reach 150°C ? 1
- iv. What would be the limiting temperature which could be achieved by the cake? 1

(c)



P and Q are the centres of the circles in the diagram above. PAX and QAY are straight lines. If $\angle PAY = x$, prove that P, Q, X and Y are concyclic.

3

End of Question 3

Question 4 (12 marks) Start a new sheet of writing paper. Marks

- (a) The acute angle between the lines L_1 and L_2 is $\frac{\pi}{4}$ radians. The equation of L_1 is $y = 3x - 1$. The equation of L_2 is $y = mx + b$. Find the equation of the line L_2 if it passes through $(-1, -4)$. 3
- (b) The equation $e^x - x - 2 = 0$ has a root close to $x = 1.2$. Use Newton's method **once** to find a better approximation to this root, correct to 2 decimal places. 2
- (c) Use mathematical induction to prove that for all positive integers n : 3
- $$4(1^3 + 2^3 + 3^3 + \dots + n^3) = n^2(n+1)^2.$$
- (i) Hence, or otherwise, find the value of: 2
- $$\lim_{n \rightarrow \infty} \left(\frac{1^3 + 2^3 + 3^3 + \dots + n^3}{n^4} \right)$$
- (d) Show that the chord of contact on the parabola $x^2 = 4ay$ is a focal chord if the external point lies on the directrix. The equation of the chord of contact is $xx_1 = 4a(y + y_1)$. Do NOT prove this result. 2

End of Question 4

Question 5 (12 marks) Start a new sheet of writing paper. Marks

- (a) i. If $g(x) = e^{x+1}$, find $g^{-1}(x)$, the equation of the inverse of $g(x)$. 2
- ii. State the domain of $g^{-1}(x)$. 1
- iii. On a number plane, sketch the graphs of $y = g(x)$ and $y = g^{-1}(x)$, showing intercepts and at least one other point on each curve. 3
- iv. Using the graphs in iii, or otherwise, discuss the symmetry of the functions $y = g(x)$ and $y = g^{-1}(x)$. 1
- (b) Air is pumped into a spherical balloon at a constant rate of $12 \text{ cm}^3/\text{s}$. Find the rate of increase in its surface area when its radius is 8 cm . 3
- (c) Find the general solution of $\cos(2x - \frac{\pi}{4}) = 1$ 2

End of Question 5

Question 6 (12 marks)

Start a new sheet of writing paper.

Marks

- (a) Find the greatest coefficient in $(5 + 2x)^{12}$. 3
- (b) An object is projected from the top of a vertical cliff 18 m above the horizontal ground at an angle θ where $\tan \theta = \frac{3}{4}$, with an initial speed of 25 m/s.
- i. Show that the equations of motion of the object are: 4
- $$x = 20t$$
- $$y = -\frac{1}{2}gt^2 + 15t + 18$$
- Neglecting air resistance, and taking $g = 9.8 \text{ m/s}^2$
- ii. What is the greatest height reached by the particle (correct to 2 decimal places)? 2
- iii. Find the distance from the base of the cliff to where the object hits the ground (correct to 2 decimal places). 3

End of Question 6**Question 7 (12 marks)**

Start a new sheet of writing paper.

Marks

- (a) A particle moves in a straight line and its position at time t is given by:
 $x = 1 + \sqrt{3} \cos 2t - \sin 2t$
- i. Show that $\sqrt{3} \cos 2t - \sin 2t = 2 \cos(2t + \frac{\pi}{6})$. 2
- ii. Show that the particle with equation $x = 1 + \sqrt{3} \cos 2t - \sin 2t$ is undergoing simple harmonic motion. 3
- iii. Describe the motion of the particle including the centre, amplitude and period of motion. 2
- iv. Find the first time the particle is at the origin (i.e. when $x = 0$). 2
- (b) Given the fact that 3
- $$\int_0^{\pi} \sin mx \sin nx \, dx = 0 \text{ and } \int_0^{\pi} \sin^2 mx \, dx = \frac{\pi}{2}$$
- if m and n are any unequal positive integers,
- find the volume obtained when the area between the curve
- $$y = \sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x$$
- and the x -axis from $x=0$ to $x=\pi$, is rotated about the x -axis.

End of Examination

PLC Sydney Maths Department

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2008
Course	Ext. 1.	Name of task/exam	Trial

Ver 1

Question 1:

$$a) \lim_{x \rightarrow 0} \frac{3 \sin \frac{x}{2}}{x} = 3 \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{x}$$

$$= 3 \cdot \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}}$$

$$= \frac{3}{2} (1)$$

$$= \frac{3}{2}$$

$$b) A \begin{pmatrix} x_1 & y_1 \\ -2 & 3 \end{pmatrix} \quad B \begin{pmatrix} x_2 & y_2 \\ 1 & 0 \end{pmatrix} \quad m:n = 4:5$$

$$x = \frac{mx_2 + nx_1}{m+n}, \quad y = \frac{my_2 + ny_1}{m+n}$$

$$x = \frac{4(1) + 5(-2)}{9}, \quad y = \frac{4(0) + 5(3)}{9}$$

$$x = -\frac{6}{9}, \quad y = \frac{15}{9}$$

$$P\left(-\frac{2}{3}, \frac{5}{3}\right)$$

$$c) x^{2k+3} = e^{9 \ln x}$$

$$x^{2k+3} = e^{\ln x^9}$$

$$\therefore 2k+3 = 9$$

$$2k = 6$$

$$k = 3$$

$$d) \int \cos^2 4x \, dx \quad \cos 8x = 2\cos^2 4x - 1$$

$$\frac{1}{2}(\cos 8x + 1) = \cos^2 4x$$

$$= \int \left(\frac{1}{2} \cos 8x + \frac{1}{2}\right) dx$$

$$= \frac{1}{2} \frac{\sin 8x}{8} + \frac{1}{2} x + c$$

$$= \frac{1}{16} \sin 8x + \frac{1}{2} x + c$$

$$e) u = 1+x^5 \quad \text{when } x = -1 \quad u = 0$$

$$\frac{du}{dx} = 5x^4 \quad \text{when } x = 1 \quad u = 2$$

$$du = 5x^4 dx$$

$$\therefore \int_{-1}^1 x^4 \sqrt{1+x^5} \, dx = \frac{1}{5} \int_0^2 5x^4 (1+x^5)^{\frac{1}{2}} dx$$

$$= \frac{1}{5} \int_0^2 u^{\frac{1}{2}} du$$

$$= \frac{1}{5} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_0^2$$

$$= \frac{2}{15} \left[u^{\frac{3}{2}} \right]_0^2$$

$$= \frac{2}{15} \left[2^{\frac{3}{2}} \right]$$

$$= \frac{4\sqrt{2}}{15}$$

Question 2:

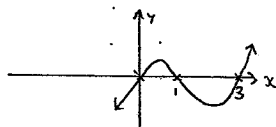
$$a) \frac{2}{x-1} \geq \frac{3}{x}, \quad x \neq 0, x \neq 1$$

$$x^2(x-1) \geq 3(x-1)^2 x$$

$$3(x-1)^2 x - 2(x-1)x^2 \leq 0$$

$$x(x-1)[3(x-1)-2x] \leq 0$$

$$x(x-1)(x-3) \leq 0$$



$$x < 0, \quad 1 < x \leq 3$$

PLC Sydney Maths Department

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2008
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Ver 1

$$\text{OR: } \frac{2}{x-1} \geq \frac{3}{x}, \quad x \neq 0, x \neq 1$$

Critical points: $x = 0, x = 1$

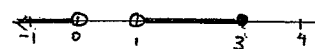
$$\frac{2}{x-1} = \frac{3}{x}$$

$$2x = 3x - 3$$

$$-x = -3$$

$$x = 3$$

$\therefore$ Look at intervals



$$\text{Test: } x = \frac{1}{2}$$

$$\frac{2}{\frac{1}{2}-1} \geq \frac{3}{\frac{1}{2}}$$

$$-4 \geq 6$$

$$\text{Test: } x = 2$$

$$\frac{2}{2-1} \geq \frac{3}{2}$$

$$2 \geq \frac{3}{2} \quad \text{True}$$

$$\text{Test: } x = 4$$

$$\frac{2}{3} \not\geq \frac{3}{4}$$

$$\text{Test: } x = -1$$

$$\frac{2}{-2} \geq \frac{3}{-1}$$

$$-1 \geq -3 \quad \text{True}$$

$$\therefore x < 0, \quad 1 < x \leq 3$$

$$2x^3 - kx + 8 = 0$$

b) Let the roots be α, α and β .

Sum of roots one at a time

$$\alpha + \alpha + \beta = -\frac{b}{a}$$

$$2\alpha + \beta = 0$$

$$\beta = -2\alpha$$

Sum of roots two at a time

$$\alpha^2 + \alpha\beta + \alpha\beta = -\frac{k}{2}$$

$$\alpha^2 + 2\alpha\beta = -\frac{k}{2}$$

$$\alpha(\alpha + 2\beta) = -\frac{k}{2}$$

$$\alpha(\alpha + 2[-2\alpha]) = -\frac{k}{2}$$

$$\alpha(\alpha - 4\alpha) = -\frac{k}{2}$$

$$\alpha(-3\alpha) = -\frac{k}{2}$$

$$3\alpha^2 = \frac{k}{2}$$

$$\alpha^2 = \frac{k}{6}$$

Product of roots

$$\alpha^2 \beta = -4$$

$$\alpha^2(-2\alpha) = -4$$

$$-2\alpha^3 = -4$$

$$\alpha^3 = 2$$

$$\alpha = \sqrt[3]{2}$$

$$\therefore k = 6\alpha^2$$

$$= 6(2^{\frac{2}{3}})$$

$$= 6(2^{\frac{2}{3}}) \quad \text{or} \quad \frac{12}{\sqrt[3]{2}}$$

$$c) T_{k+1} = {}^8C_k (x^3)^k \left(-\frac{1}{x}\right)^{8-k}$$

$$= {}^8C_k x^{3k} (-1)^{8-k} x^{k-8}$$

$$= {}^8C_k (-1)^{8-k} x^{4k-8}$$

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for constant term: $4k - 8 = 0$

$$\therefore k = 2.$$

Constant term is $8C_2(-1)^6$
 $= 28$

d) $\ddot{x} = -8x^3$

when $t=0$, $x=0$, $v=4 \text{ m/s}$

1 max speed is when $\ddot{x} = 0$

$$\text{i.e. } 0 = -8x^3$$

$$x = 0.$$

So when particle is at origin it has max speed.

$\therefore$ max speed = 4 m/s .

ii) $\frac{d}{dx}(\frac{1}{2}v^2) = -8x^3$

$$\frac{1}{2}v^2 = -8 \int x^3 dx$$

$$\frac{1}{2}v^2 = -8 \frac{x^4}{4} + c$$

$$\frac{1}{2}v^2 = -2x^4 + c$$

$$\text{when } x=0 \text{ } v=4$$

$$8 = 0 + c$$

$$\therefore \frac{1}{2}v^2 = -2x^4 + 8$$

$$v^2 = -4x^4 + 16$$

$$v^2 = 4(4 - x^4)$$

$$v = \pm \sqrt{4(4 - x^4)}$$

$$= \pm 2(\sqrt{4 - x^4})$$

$$\text{Since } v=4 \text{ when } x=0$$

$$\therefore v = 2\sqrt{4 - x^4}$$

Question 3:

a) i) Show $\cos 3x = 4\cos^3 x - 3\cos x$

$$\text{LHS} = \cos(2x+x)$$

$$= \cos 2x \cos x - \sin 2x \sin x$$

$$= (2\cos^2 x - 1)\cos x - 2\sin x \cos x \sin x$$

$$= 2\cos^3 x - \cos x - 2\sin^2 x \cos x$$

$$= 2\cos^3 x - \cos x - 2(1 - \cos^2 x)\cos x$$

$$= 2\cos^3 x - \cos x - 2\cos x + 2\cos^3 x$$

$$= 4\cos^3 x - 3\cos x$$

$$= \text{RHS}$$

ii) $\int \cos x \sin^2 x dx$

Method 1:

$$\int \cos x (1 - \cos^2 x) dx$$

$$= \int (\cos x - \cos^3 x) dx$$

$$= \int \left(\cos x - \left[\frac{1}{4}(\cos 3x + 3\cos x) \right] \right) dx$$

$$= \int \left(\cos x - \frac{1}{4}\cos 3x - \frac{3}{4}\cos x \right) dx$$

$$= \int \left(\frac{1}{4}\cos x - \frac{1}{4}\cos 3x \right) dx$$

$$= \frac{1}{4}\sin x - \frac{1}{12}\sin 3x + c$$

Method 2:

$$\int f'(x) [f(x)]^n dx$$

$$= \frac{[f(x)]^{n+1}}{n+1} + c$$

$$\therefore \int \cos x (\sin x)^2 dx$$

$$= \frac{\sin^3 x}{3} + c$$

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Method 3 Let $u = \sin x$

$$\frac{du}{dx} = \cos x$$

$$du = \cos x dx$$

$$\therefore \int \cos x \sin^2 x dx = \int u^2 du$$

$$= \frac{u^3}{3} + c$$

$$= \frac{\sin^3 x}{3} + c.$$

Note: $\frac{1}{4}\sin x - \frac{1}{12}\sin 3x$

$$= \frac{1}{4}\sin x - \frac{1}{12}[\sin(2x+x)]$$

$$= \frac{1}{4}\sin x - \frac{1}{12}[\sin 2x \cos x + \cos 2x \sin x]$$

$$= \frac{1}{4}\sin x - \frac{1}{12}[2\sin x \cos^2 x + (1 - 2\sin^2 x)\sin x]$$

$$= \frac{1}{4}\sin x - \frac{1}{12}[2\sin x(1 - \sin^2 x) + \sin x - 2\sin^3 x]$$

$$= \frac{1}{4}\sin x - \frac{2}{12}\sin x + \frac{2}{12}\sin^3 x - \frac{1}{12}\sin x + \frac{2}{12}\sin^3 x$$

$$= \frac{1}{3}\sin^3 x$$

b) i) $\frac{dT}{dt} = k(N - P)$

$$T = P + Ae^{kt}$$

$$P = 180$$

$$\text{when } t=0 \text{ } T=20$$

$$t=0 \text{ } T=20$$

$$t=15 \text{ } T=100$$

$$20 = 180 + Ae^0$$

$$A = -160$$

$$T = 180 - 160e^{kt}$$

$$\text{when } t=15 \text{ } T=100$$

$$100 = 180 - 160e^{15k}$$

$$-80 = -160e^{15k}$$

$$\frac{1}{2} = e^{15k}$$

$$\ln \frac{1}{2} = 15k$$

$$k = \frac{1}{15} \ln \frac{1}{2}$$

$$\therefore k = -0.0462098 \dots$$

$$T = 180 - 160e^{-0.046t}$$

ii) when $t=30$ $T=?$

$$T = 180 - 160e^{-0.046 \times 30}$$

$$T = 140$$

$\therefore$ temp will be 140°C

iii) $T=150$ $t=?$

$$150 = 180 - 160e^{-0.046 \dots t}$$

$$-30 = -160e^{-0.046 \dots t}$$

$$\frac{-30}{-160} = e^{-0.046 \dots t}$$

$$\ln \frac{3}{16} = -0.046 \dots t$$

$$t = 36 \text{ mins } 13.5 \text{ sec.}$$

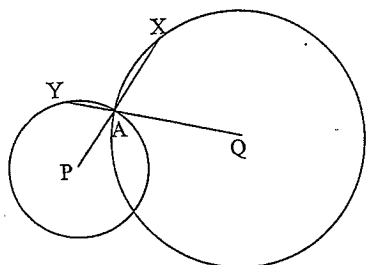
iv) $T = 180 - 160e^{-0.046t}$

$$\text{as } t \rightarrow \infty$$

limiting temp is 180°C .

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2008
Course	Ext. 1	Name of task/exam	Trial

IF $\angle PAY = x$ then $\angle QAX = x$ (vertically opposite angles are equal)

Join PY and QX

 $PA = PY$ (equal radii) $QA = QX$ (equal radii) $\therefore \angle PYA = \angle PAY = x$ (angles opposite equal sides are equal)

and

 $\angle QAX = \angle QXA$ (angles opposite equal sides are equal) $\therefore \angle PYA = \angle QXA$ $\therefore PQXY$ are concyclic as angles subtended on the same side of chord, PQ, are equal.OR IF $\angle PAY = x$ then $\angle QAX = x$ (vertically opposite angles are equal)

Join PY and QX

 $PA = PY$ (equal radii) $QA = QX$ (equal radii) $\therefore \angle PYA = \angle PAY = x$ (angles opposite equal sides are equal)and $\angle QAX = \angle QXA = x$ (similarly) $\therefore \triangle APY \parallel \triangle AQX$ (equiangular) $\therefore$ In similar triangles, corresponding sides are in the same ratio

$$\text{i.e. } \frac{PA}{QA} = \frac{YA}{XA}$$

$$\therefore PA \times XA = YA \times QA$$

Since the product of the intercepts on intersecting intervals is equal then the endpoints of the intervals are concyclic

Question 4:

$$a. y = 3x - 1 \quad m_1 = 3$$

$$y = mx + b \quad m_2 = m$$

$$\therefore \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \frac{\pi}{4} = \left| \frac{3 - m}{1 + m(3)} \right|$$

$$\pm 1 = \frac{3 - m}{1 + 3m}$$

$$1 + 3m = 3 - m \quad -1 - 3m = 3 - m$$

$$4m = 2 \quad -4 = 2m$$

$$m = \frac{1}{2} \quad m = -2$$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2008
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 $\therefore$ eqn of line

$$y + 4 = \frac{1}{2}(x + 1) \quad y + 4 = -2(x + 1)$$

$$2y + 8 = x + 1 \quad y + 4 = -2x - 2$$

$$x - 2y - 7 = 0 \quad 2x + y + 6 = 0$$

$$\text{or } y = \frac{1}{2}x - 3\frac{1}{2} \quad \text{or } y = -2x - 6$$

b. Let $P(x) = e^x - x - 2$

$$P(1.2) = 0.1201169...$$

$$P'(x) = e^x - 1$$

$$P'(1.2) = 2.3201169...$$

$$\therefore x_2 = x_1 - \frac{P(x_1)}{P'(x_1)}$$

$$= 1.2 - \frac{0.1201169...}{2.3201169...}$$

$$= 1.148...$$

$$= 1.15$$

 $\therefore$ a better approximation is 1.15 and $P(x)$ is continuous.

c. i

Prove

$$4(1^3 + 2^3 + 3^3 + \dots + n^3) = n^2(n+1)^2$$

Step 1: prove true for $n=1$

$$\text{LHS} = 4(1^3) \quad \text{RHS} = 1^2(1+1)^2$$

$$= 4 \quad = 1(4) = 4$$

 $\therefore$ true for $n=1$ Step 2: assume true for $n=k$

$$\text{i.e. } 4(1^3 + 2^3 + 3^3 + \dots + k^3) = k^2(k+1)^2$$

Step 3: prove true for $n=k+1$

i.e. prove

$$4(1^3 + 2^3 + \dots + k^3 + (k+1)^3) = (k+1)^2(k+2)^2$$

$$\text{LHS} = 4(1^3 + 2^3 + \dots + k^3) + 4(k+1)^3$$

$$= k^2(k+1)^2 + 4(k+1)^3 \text{ from assumpt.}$$

$$= (k+1)^2[k^2 + 4(k+1)]$$

$$= (k+1)^2[k^2 + 4k + 4]$$

$$= (k+1)^2(k+2)^2$$

$$= \text{RHS}$$

 $\therefore$ true $\therefore$ By the principle of mathematical induction it is true for all positive integers n .

$$\text{ii. } \lim_{x \rightarrow \infty} \frac{(1^3 + 2^3 + 3^3 + \dots + n^3)}{n^4}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{n^2(n+1)^2}{4}}{n^4}$$

$$= \lim_{x \rightarrow \infty} \frac{n^2(n+1)^2}{4n^4}$$

$$= \lim_{x \rightarrow \infty} \frac{n^2(n^2 + 2n + 1)}{4n^4}$$

$$= \lim_{x \rightarrow \infty} \frac{n^4 + 2n^3 + n^2}{4n^4}$$

$$= \frac{1}{4}$$

Solutions for exams and assessment tasks

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$$d. \quad x x_1 = 4a(y+y_1)$$

If external point lies on directrix, coordinates would be $(x_1, -a)$

$$\therefore x x_1 = 4a(y + -a)$$

$$x x_1 = 4a(y - a)$$

If a focal chord $(0, a)$ satisfies

$$\text{LHS} = 0(x_1) \quad \text{RHS} = 4a(a - a)$$

$$= 0 \quad = 0$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore x x_1 = 4a(y - a)$ is a focal chord.

If external point lies on directrix, the chord of contact is a focal chord.

Question 5:

$$a. \quad i. \quad g(x) = e^{x+1}$$

$$\text{i.e. } y = e^{x+1}$$

$$\text{inverse: } x = e^{y+1}$$

$$\ln x = y + 1$$

$$y = (\ln x) - 1$$

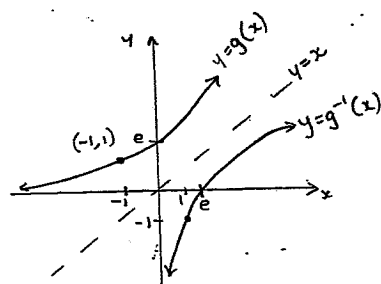
$$\therefore g^{-1}(x) = -1 + \ln x$$

ii. domain of $g^{-1}(x)$ = range of $g(x)$

$\therefore$ range of $g(x)$ is $y > 0$

$\therefore$ domain of $g^{-1}(x)$: $x > 0$

iii



iv. The graphs are symmetrical about the line $y = x$.

$$b. \quad V_{\text{sphere}} = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\frac{dV}{dt} = 12 \text{ cm}^3/\text{s}$$

$$SA_{\text{sphere}} = 4\pi r^2$$

$$\frac{dS}{dr} = 8\pi r$$

$$\frac{dS}{dt} = ? \quad \text{when } r = 8$$

$$\frac{dS}{dt} = \frac{dS}{dr} \times \frac{dr}{dt} \times \frac{dr}{dV}$$

$$\frac{dS}{dt} = 8\pi r \times 12 \times \frac{1}{4\pi r^2}$$

$$= \frac{24}{r}$$

$$\text{when } r = 8$$

$$\frac{dS}{dt} = 3 \text{ cm}^2/\text{s}$$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2008
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OR

$$\frac{dV}{dt} = 12 \quad \frac{dS}{dt} = ? \quad \text{when } r = 8$$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$12 = 4\pi r^2 \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{12}{4\pi r^2}$$

$$\frac{dr}{dt} = \frac{3}{\pi r^2}$$

$$S = 4\pi r^2$$

$$\frac{dS}{dr} = 8\pi r$$

$$\frac{dS}{dt} = \frac{dS}{dr} \times \frac{dr}{dt}$$

$$\frac{dS}{dt} = \frac{8\pi r}{1} \times \frac{3}{\pi r^2}$$

$$\frac{dS}{dt} = \frac{24}{r}$$

$$\text{when } r = 8$$

$$\frac{dS}{dt} = \frac{24}{8}$$

$$\frac{dS}{dt} = 3 \text{ cm}^2/\text{sec}$$

$$c. \quad \cos\left(2x - \frac{\pi}{4}\right) = 1$$

$$2x - \frac{\pi}{4} = 2\pi n$$

$$2x = 2\pi n + \frac{\pi}{4}$$

$$x = \pi n + \frac{\pi}{8}, \quad n \text{ integer}$$

Question 6:

$$a. \quad (5 + 2x)^{12}$$

$$T_{k+1} = {}^{12}C_k (5)^k (2x)^{12-k}$$

$$T_k = {}^{12}C_{k-1} (5)^{k-1} (2x)^{12-(k-1)}$$

Coeff:

$$\frac{T_{k+1}}{T_k} = \frac{{}^{12}C_k 5^k 2^{12-k}}{{}^{12}C_{k-1} 5^{k-1} 2^{13-k}}$$

$$= \frac{12!}{k!(12-k)!} \cdot \frac{5^k 2^{12-k}}{5^{k-1} 2^{13-k}}$$

$$\frac{T_{k+1}}{T_k} = \frac{12!}{k!(12-k)!} \times \frac{(k-1)!(13-k)!}{5^{k-1} 2^{13-k}}$$

$$= 13 - k \times \frac{5}{2}$$

$$= \frac{65 - 5k}{2}$$

$$\text{for greatest coefficient } \frac{T_{k+1}}{T_k} > 1$$

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$$\frac{65-5k}{2k} > 1$$

$$65-5k > 2k$$

$$-7k > -65$$

$$k < \frac{65}{7}$$

$$k < 9\frac{2}{7}$$

$$\therefore k = 9$$

Greatest Coefficient is

$${}^{12}C_9 5^9 2^3 = 3437500000$$

OR: $(5+2x)^{12}$

$$T_{k+1} = {}^{12}C_k (2x)^k (5)^{12-k}$$

$$T_k = {}^{12}C_{k-1} (2x)^{k-1} 5^{12-(k-1)}$$

Coeff:

$$\frac{T_{k+1}}{T_k} = \frac{{}^{12}C_k 2^k 5^{12-k}}{{}^{12}C_{k-1} 2^{k-1} 5^{13-k}}$$

$$= \frac{12-k+1}{k} \times \frac{2}{5}$$

$$= \frac{(13-k)2}{5k}$$

for greatest coefficient $\frac{T_{k+1}}{T_k} > 1$

$$\therefore \frac{2(13-k)}{5k} > 1$$

$$26-2k > 5k$$

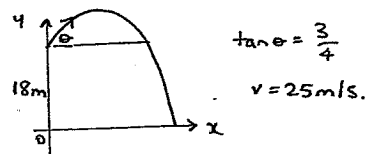
$$26 > 7k$$

$$k < 3\frac{5}{7}$$

Greatest coefficient is

$${}^{12}C_3 2^3 5^9 = 3437500000$$

b.



Initially

$$\cos \theta = \frac{x}{25}$$

$$\dot{x} = 25 \cos \theta$$

$$\sin \theta = \frac{y}{25}$$

$$\dot{y} = 25 \sin \theta$$

we know $\tan \theta = \frac{3}{4}$



$$\therefore \cos \theta = \frac{4}{5}$$

$$\sin \theta = \frac{3}{5}$$

$$\therefore \dot{x} = 25 \times \frac{4}{5}$$

$$= 20$$

$$\dot{y} = 25 \times \frac{3}{5}$$

$$= 15$$

$$\ddot{x} = 0$$

$$\dot{x} = \int dt$$

$$\dot{x} = C_1$$

$$20 = C_1$$

$$\therefore \dot{x} = 20$$

$$x = \int 20 dt$$

$$x = 20t + C_2$$

when $x = 0$ $t = 0$

$$\therefore C_2 = 0$$

$$\therefore x = 20t$$

Academic Year	Yr 12	Calendar Year	2008
Course	Ext. 1	Name of task/exam	Trial

$$\ddot{y} = -g$$

$$\dot{y} = \int -g dt$$

$$\dot{y} = -gt + C_3$$

when $t = 0$ $\dot{y} = 15$

$$15 = C_3$$

$$\therefore \dot{y} = -gt + 15$$

$$y = \int (-gt + 15) dt$$

$$y = -\frac{1}{2}gt^2 + 15t + C_4$$

when $t = 0$ $y = 18$

$$18 = C_4$$

$$\therefore y = -\frac{1}{2}gt^2 + 15t + 18$$

ii. greatest height is when

$$\dot{y} = 0$$

$$\dot{y} = -gt + 15$$

$$0 = -9.8t + 15$$

$$9.8t = 15$$

$$t = \frac{15}{9.8}$$

$$= 1.5306 \dots$$

when $t = 1.53 \dots$

$$y = ?$$

$$y = -\frac{1}{2}(9.8)(1.53 \dots)^2 + 15(1.53 \dots) + 18$$

$$= 29.47959 \dots$$

$\therefore$ greatest height is 29.48 m.

iii. when $y = 0$ $t = ?$

$$0 = -\frac{1}{2}gt^2 + 15t + 18$$

$$t = \frac{-15 \pm \sqrt{225 - 4(-\frac{1}{2}g)(18)}}{2(-\frac{1}{2}g)}$$

$$t = 3.9834 \dots \quad t > 0$$

$$\therefore x = 20t$$

$$\therefore x = 20(3.9834 \dots)$$

$$x = 79.668 \dots$$

$\therefore$ distance is 79.67 m

Question 7:

$$a. \downarrow \sqrt{3} \cos 2t - \sin 2t = R \cos(2t + \alpha)$$

$$\sqrt{3} \cos 2t - \sin 2t = R \cos 2t \cos \alpha - R \sin 2t \sin \alpha$$

$$\sqrt{3} = R \cos \alpha \quad (1)$$

$$1 = R \sin \alpha \quad (2)$$

$$4 = R^2 (\cos^2 \alpha + \sin^2 \alpha) \quad (1)^2 + (2)^2$$

$$R = 2 \quad R > 0$$

$$\tan \alpha = \frac{1}{\sqrt{3}} \quad (2) \div (1)$$

$$\alpha = \frac{\pi}{6}$$

$$\therefore \sqrt{3} \cos 2t - \sin 2t = 2 \cos(2t + \frac{\pi}{6})$$

OR

$$RHS = 2 \cos(2t + \frac{\pi}{6})$$

$$= 2 \left[\cos 2t \cos \frac{\pi}{6} - \sin 2t \sin \frac{\pi}{6} \right]$$

$$= 2 \left[\cos 2t \left(\frac{\sqrt{3}}{2} \right) - \sin 2t \left(\frac{1}{2} \right) \right]$$

$$= \sqrt{3} \cos 2t - \sin 2t$$

$$= LHS$$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2008
Course	Ext 1	Name of task/exam	Trial

$$ii \quad x = 1 + \sqrt{3} \cos 2t - \sin 2t$$

$$x = 1 + 2 \cos(2t + \frac{\pi}{6})$$

$$x - 1 = 2 \cos(2t + \frac{\pi}{6})$$

$$\dot{x} = -4 \sin(2t + \frac{\pi}{6})$$

$$\ddot{x} = -8 \cos(2t + \frac{\pi}{6})$$

$$\ddot{x} = -4 [2 \cos(2t + \frac{\pi}{6})]$$

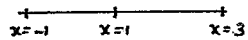
$$\ddot{x} = -4 [x - 1]$$

$\therefore$ SHM.

iii Centre is at $x = 1$

amplitude is 2

period is π sec.



$$iv \quad x = 1 + 2 \cos(2t + \frac{\pi}{6})$$

when $x = 0$

$$-1 = 2 \cos(2t + \frac{\pi}{6})$$

$$-\frac{1}{2} = \cos(2t + \frac{\pi}{6})$$

$$2t + \frac{\pi}{6} = \frac{2\pi}{3}, \dots$$

$$2t = \frac{2\pi}{3} - \frac{\pi}{6}$$

$$2t = \frac{\pi}{2}$$

$$t = \frac{\pi}{4} \text{ sec}$$

$$b \quad V = \pi \int_0^{\pi} \left[\left(\sin x + \frac{1}{3} \sin 3x \right) + \frac{1}{5} \sin 5x \right]^2 dx$$

$$= \pi \int_0^{\pi} \left[\left(\sin x + \frac{1}{3} \sin 3x \right)^2 + \frac{2}{5} \sin 5x \left(\sin x + \frac{1}{3} \sin 3x \right) + \frac{1}{25} \sin^2 5x \right] dx$$

$$= \pi \int_0^{\pi} \left(\sin^2 x + \frac{2}{3} \sin x \sin 3x + \frac{1}{9} \sin^2 3x \right. \\ \left. + \frac{2}{5} \sin x \sin 5x + \frac{2}{15} \sin 5x \sin 3x \right. \\ \left. + \frac{1}{25} \sin^2 5x \right) dx$$

$$\text{Since } \int_0^{\pi} \sin mx \sin nx dx = 0$$

$$\therefore V = \pi \int_0^{\pi} \left(\sin^2 x + 0 + \frac{1}{9} \sin^2 3x + 0 + 0 + \frac{1}{25} \sin^2 5x \right) dx$$

$$\text{Since } \int_0^{\pi} \sin^2 mx dx = \frac{\pi}{2}$$

$$\therefore V = \pi \left[\frac{\pi}{2} + \frac{1}{9} \left(\frac{\pi}{2} \right) + \frac{1}{25} \left(\frac{\pi}{2} \right) \right]$$

$$= \pi^2 \left[\frac{1}{2} + \frac{1}{18} + \frac{1}{50} \right]$$

$$= \frac{259\pi^2}{450} \text{ units}^3$$